# IRC Results — Full Sella_hip Sweep

sella_hip IRC on gad_dt003 converged TSs, 2026-04-16

## Scope

Full IRC validation of the best GAD variant (`gad_dt003`, projected GAD with fixed $\mathrm{d}t = 0.003$) against the `Transition1x` train split reactant/product labels using `sella_hip` — Sella's IRC algorithm with the BFGS Hessian update replaced by HIP's analytic mass-weighted, Eckart-projected Hessian after every inner step.

**Pipeline.**

1. Start from a `Transition1x` labeled TS, perturb by isotropic Gaussian noise (6 noise levels: 10/30/50/100/150/200
2. Run `gad_dt003`; at the first step where Eckart-projected $n_{\mathrm{neg}} = 1$ and $\|\mathbf{F}\| < 0.01\,\mathrm{eV/Å}$, take the trajectory coordinates at that step. No refinement, no re-gating.
3. From that TS, run `sella_hip` IRC forward/reverse for up to 500 steps.
4. Compare forward/reverse endpoints to labeled reactant/product using element-aware Kabsch+Hungarian RMSD and bond-graph isomorphism.
5. Re-evaluate HIP's vibrational Hessian at both endpoints to confirm each is a true minimum ($n_{\mathrm{neg,vib}} = 0$).

**Primary metric.** *TOPO-intended*: both directions match (reactant, product) by bond-graph isomorphism. RMSD-intended uses the stricter $< 0.3\,\mathrm{Å}$ threshold on the same direction-pair. Both labels are direction-agnostic (forward/reverse can swap).

## Headline Numbers

1462 IRC runs across 6 noise levels, zero errors, average 11.9 s wall per sample (forward+reverse IRC).

| Noise | $n$ | TOPO-int% | TOPO-half% | RMSD-int% | RMSD-half% | $\mathrm{wall}_{avg}$ (s) |
|---|---|---|---|---|---|---|
| 10 pm | 284 | 94.4 | 5.6 | 64.4 | 32.0 | 12.6 |
| 30 pm | 283 | 94.3 | 5.6 | 64.3 | 32.5 | 12.4 |
| 50 pm | 276 | 93.1 | 6.5 | 64.9 | 31.2 | 12.3 |
| 100 pm | 262 | 93.9 | 5.7 | 65.6 | 30.5 | 12.2 |
| 150 pm | 214 | 90.7 | 9.3 | 66.8 | 29.9 | 11.8 |
| 200 pm | 143 | 80.4 | 18.2 | 60.8 | 34.3 | 10.4 |
| all | 1462 | **92.1** | 7.9 | **64.7** | 31.8 | 11.9 |

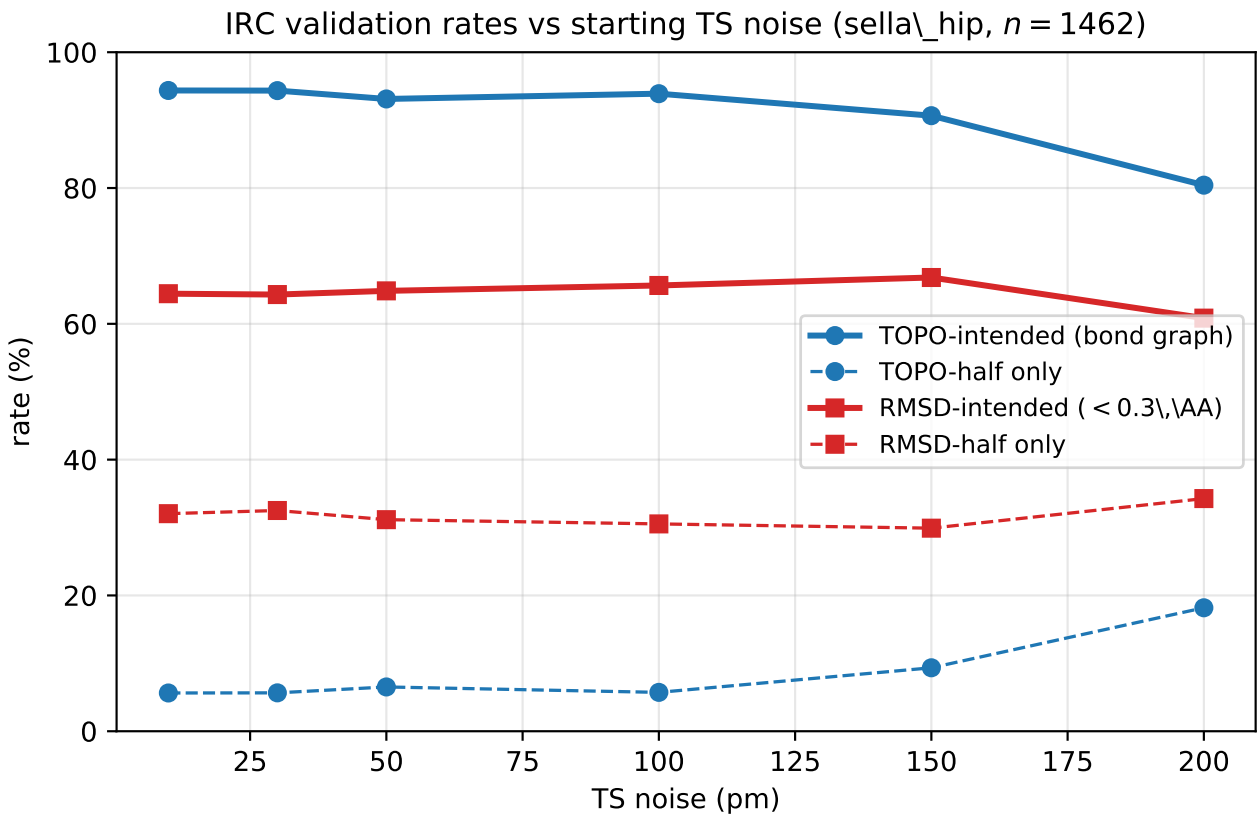


**Figure 1:** IRC validation rates vs. starting TS noise. Topology success is nearly flat at ∼93–94% through 100 pm, steps down to 90.7% at 150 pm, and drops to 80.4% at 200 pm. Strict RMSD-intended is roughly *flat or slightly rising* through 150 pm (64–67%) before dropping at 200 pm — implying the strict-RMSD failures are not noise-driven; they are conformational or labeling-driven.

## Endpoint Vibrational Quality

For each IRC endpoint we re-compute HIP's analytic Hessian, project out translation/rotation (Eckart), and inspect the vibrational spectrum. An endpoint is a *true minimum* iff $n_{\mathrm{neg,vib}} = 0$ *and* $\min \lambda_{\mathrm{vib}} > 0$.

| Noise | fwd at min% | rev at min% | both at min% | neither at min% | any-dir match†% |
|---|---|---|---|---|---|
| 10 pm | 91.5 | 95.1 | 87.7 | 1.1 | 100.0 |
| 30 pm | 91.2 | 94.7 | 86.9 | 1.1 | 100.0 |
| 50 pm | 92.0 | 93.8 | 87.7 | 1.8 | 99.6 |
| 100 pm | 93.1 | 94.7 | 88.6 | 0.8 | 99.6 |
| 150 pm | 93.5 | 95.8 | 89.7 | 0.5 | 100.0 |
| 200 pm | 93.0 | 95.8 | 89.5 | 0.7 | 98.6 |

†forward or reverse matched *some* label by bond graph.

## Topology vs RMSD Gap

Bond-graph topology is far more lenient than 0.3 Å RMSD. Among the 1347 TOPO-intended runs:

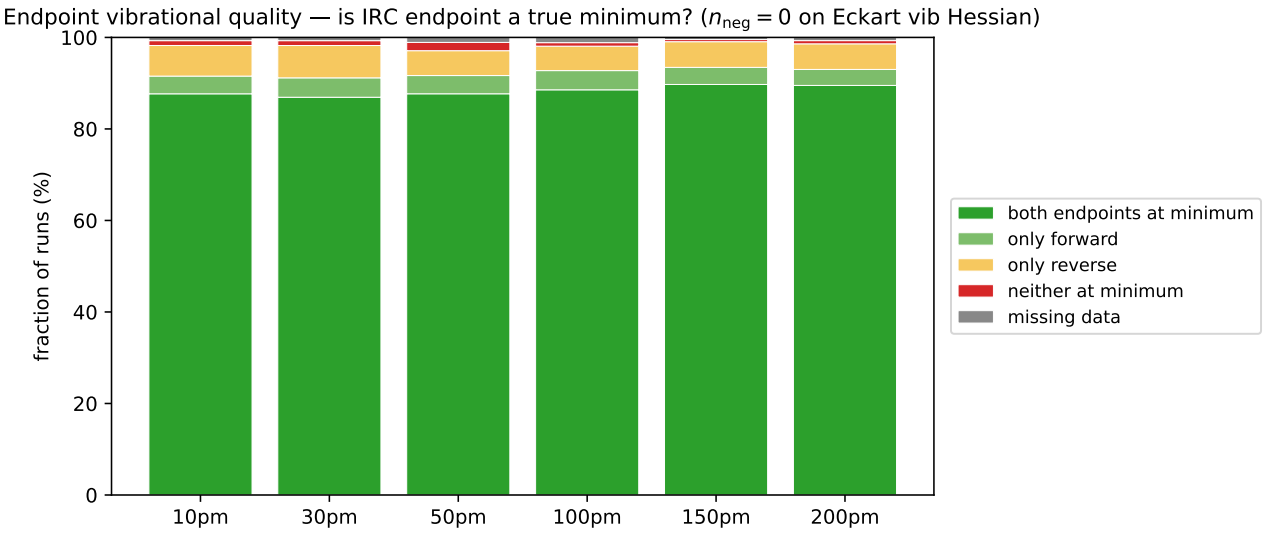


**Figure 2:** Fraction of IRC endpoints classified by vibrational quality. Both endpoints are true minima in 87–90% of runs across all noise levels. "neither at minimum" (integrator stalled on a ridge) is 0.5–1.8% — a tight upper bound on genuine IRC failure.

| Noise | TOPO-int | also RMSD-int | only TOPO | median RMSD →R (Å) |
|---|---|---|---|---|
| 10 pm | 268 | 183 | 85 | 0.009 |
| 30 pm | 267 | 182 | 85 | 0.009 |
| 50 pm | 257 | 179 | 78 | 0.009 |
| 100 pm | 246 | 172 | 74 | 0.009 |
| 150 pm | 194 | 143 | 51 | 0.008 |
| 200 pm | 115 | 87 | 28 | 0.008 |

The median RMSD to the labeled reactant *among TOPO-intended runs* is 0.008–0.009 Å. The distribution is clearly bimodal: most topology-intended runs land exactly on the label, and a smaller cluster reaches the same bond-graph via a different conformer.

## Directional Symmetry

| Noise | fwd match% | rev match% | only-fwd% | only-rev% | both% | neither% |
|---|---|---|---|---|---|---|
| 10 pm | 96.8 | 97.5 | 2.5 | 3.2 | 94.4 | 0.0 |
| 30 pm | 97.2 | 97.2 | 2.8 | 2.8 | 94.4 | 0.0 |
| 50 pm | 96.7 | 96.0 | 3.6 | 2.9 | 93.1 | 0.4 |
| 100 pm | 97.0 | 96.6 | 3.1 | 2.7 | 93.9 | 0.4 |
| 150 pm | 98.1 | 94.9 | 5.1 | 1.9 | 93.0 | 0.0 |
| 200 pm | 93.7 | 90.9 | 7.7 | 4.9 | 86.0 | 1.4 |

The forward and reverse directions hit labeled endpoints at nearly identical rates (difference ≤3 pp at every noise level except 150 pm, where forward is slightly more reliable). "Only one direction matched" is always ≤8%. This is the expected symmetry for a well-behaved IRC integrator; the HIP-injected Sella shows no directional bias.

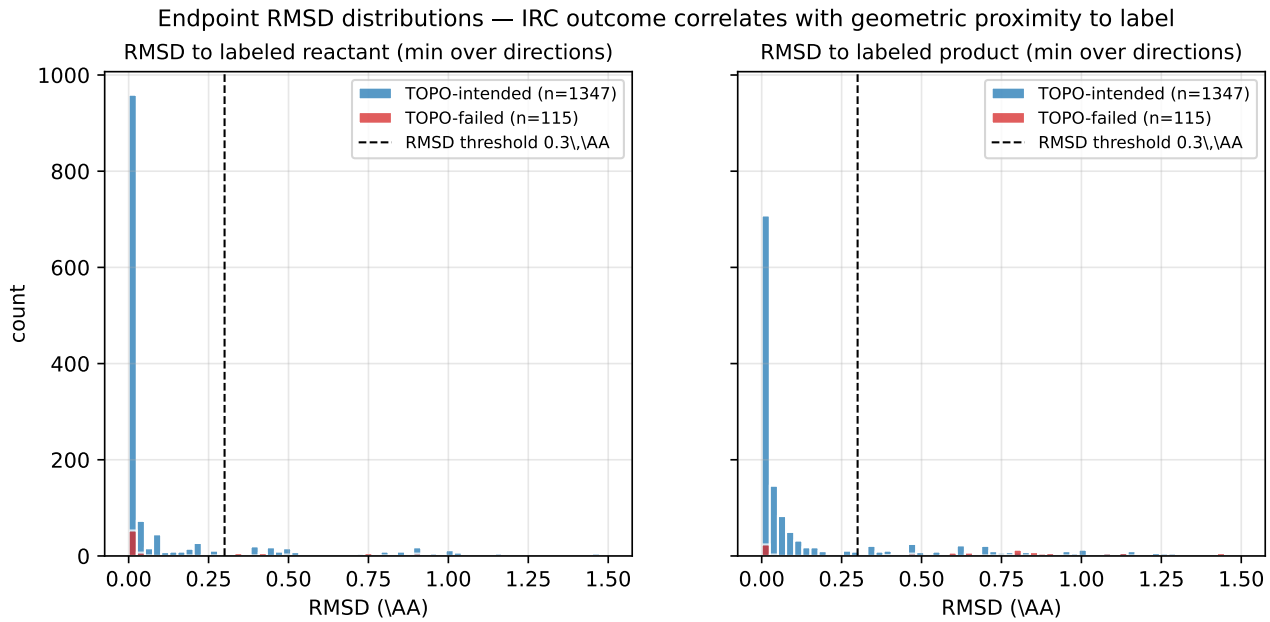


**Figure 3:** Endpoint RMSD to labeled reactant (left) and product (right), split by IRC outcome. TOPO-intended runs are tightly clustered near zero. TOPO-failed runs show a broad tail at 1–2 Å — consistent with IRC descending to a different minimum on the PES.

## Systematic Failures — Same Samples Fail at Every Noise Level

Fifteen samples are present at ≥4 noise levels and fail TOPO at every one of them. Ten of them fail at ≥5 of the 6 levels:

| sid | formula | fails/total | median best RMSD (Å) | median $[n_{\text{neg,fwd}}, n_{\text{neg,rev}}]$ | class |
|---|---|---|---|---|---|
| 48 | C2H3N3 | 6/6 | 0.00 | [0, 0] | valid-but-wrong |
| 94 | C2H3N3O | 6/6 | 0.01 | [0, 0] | valid-but-wrong |
| 96 | C2H3N3O | 6/6 | 0.01 | [0, 0] | valid-but-wrong |
| 98 | C2H3N3O | 6/6 | 0.01 | [0, 0] | valid-but-wrong |
| 104 | C2H3N3O | 6/6 | 0.02 | [0, 0] | valid-but-wrong |
| 108 | C2H3N5 | 6/6 | 0.01 | [0, 0] | valid-but-wrong |
| 143 | C2H4N2O2 | 6/6 | 0.02 | [0, 0] | valid-but-wrong |
| 203 | C2H4N4 | 6/6 | 0.02 | [0, 0] | valid-but-wrong |
| 258 | C2H6O2 | 6/6 | 0.35 | [1, 0] | ridge-stall |
| 265 | C3H2N2O | 5/5 | 0.42 | [–, 0] | partial |

Eight of the ten reach valid minima on both sides (*valid-but-wrong*): GAD converged to a real TS, IRC descended to two real minima, just not the pair the `Transition1x` dataset labels for that reaction. This is a dataset-side/labeling issue or a case where multiple reaction channels exist from the same saddle.

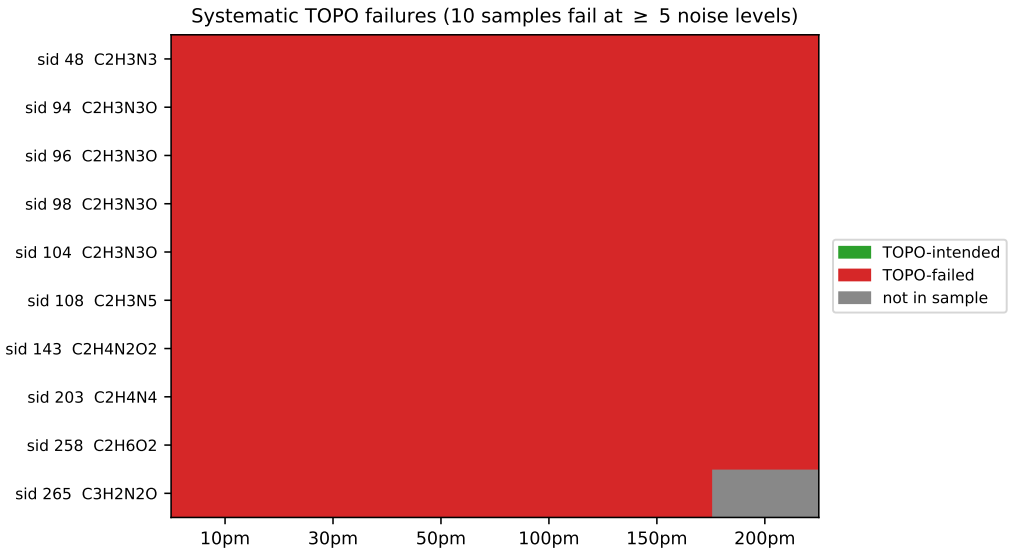


**Figure 4:** Same-sample TOPO outcome across the 6 noise levels. Persistent red rows flag systematic failures — these are reaction-labeling artifacts, not IRC-integrator stochasticity.

## TS Quality vs IRC Outcome

Cross-linking IRC outcome with the upstream GAD convergence trajectory:

| Noise | $\text{eig}_0$ median (int) | $\text{eig}_0$ median (fail) | cstep median (int) | cstep median (fail) |
|---|---|---|---|---|
| 10 pm | −3.56 | −7.01 | 95 | 107 |
| 30 pm | −3.46 | −7.03 | 206 | 220 |
| 50 pm | −3.66 | −6.65 | 295 | 321 |
| 100 pm | −4.45 | −7.02 | 454 | 469 |
| 150 pm | −6.48 | −2.30 | 585 | 886 |
| 200 pm | −8.85 | −7.64 | 662 | 1110 |

Two observations:

- At low–mid noise, TOPO-failing TSs have *sharper* saddles (more-negative $\lambda_0$) than successful ones. Interpretation: these are steeply-curved saddles where IRC escapes quickly into one basin, but the "right" product requires a more gradual descent.
- At high noise (150/200 pm), TOPO-failing TSs converge with substantially *more GAD steps* (`cstep` median +50% above successful). These are ambiguous saddles: GAD struggles to find them, and when it does, IRC often ends up in the wrong basin.

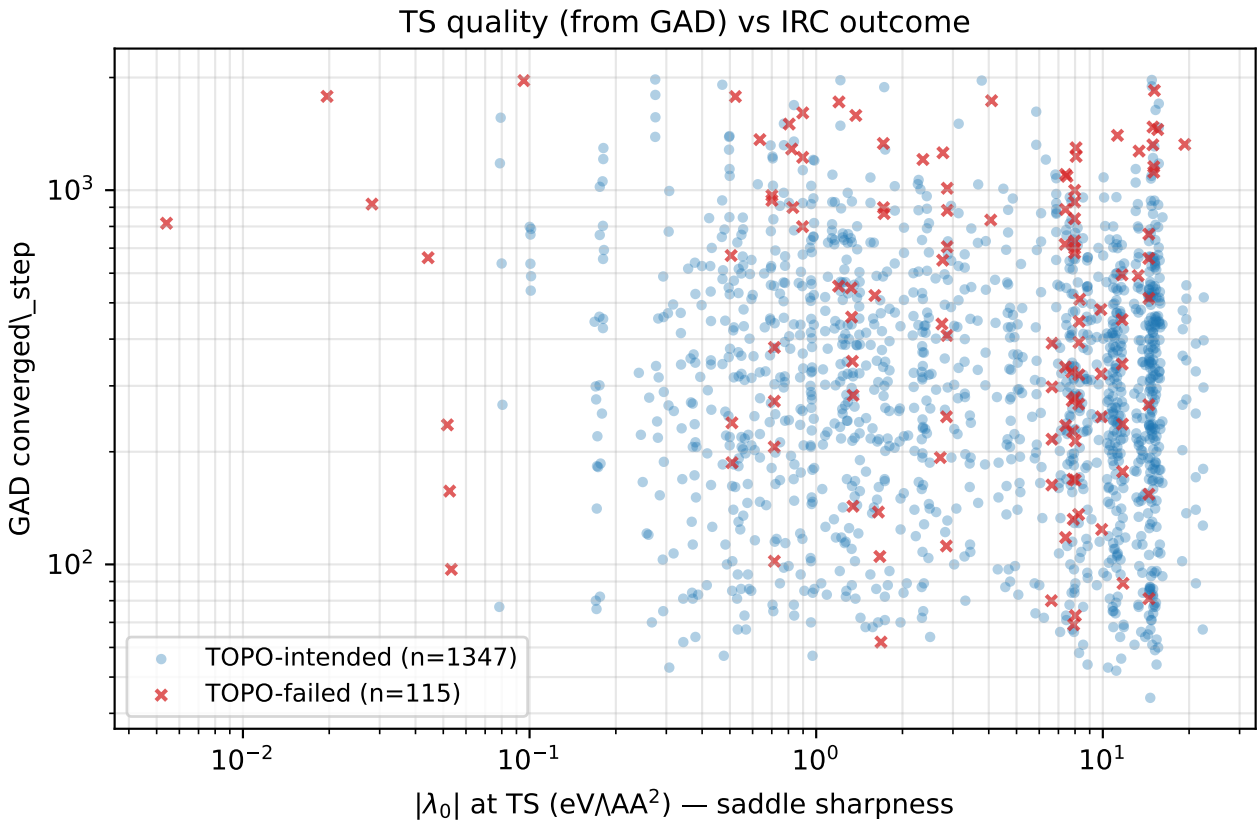


**Figure 5:** $|\lambda_0|$ at TS convergence vs the GAD step at which it converged. TOPO-failed runs are not concentrated in any single region; they spread across both axes — no simple TS-side quality gate would pre-filter them.

## Per-Formula Difficulty

| formula | $n$ | TOPO-int% | wall$_{avg}$ (s) |
|---|---|---|---|
| C2N2 | 4 | 50.0 | 19.8 |
| C3H2N2O | 15 | 60.0 | 15.1 |
| C2H3N5 | 38 | 71.1 | 18.9 |
| C2H6O2 | 26 | 76.9 | 10.6 |
| C2H4N2O | 34 | 82.4 | 9.3 |
| C2H3N3O | 306 | 88.6 | 12.3 |
| C2H4N2O2 | 58 | 89.7 | 9.9 |
| C2H5NO | 57 | 91.2 | 8.2 |
| C2H4N4 | 263 | 93.9 | 8.9 |
| C3H2N2O2 | 68 | 94.1 | 21.0 |
| 10 additional formulas are TOPO-intended at 100% ($n \geq 3$). | | | |

## Wall Time

| Noise | mean | 50% | 90% | 99% | max |
|---|---|---|---|---|---|
| 10 pm | 12.6 | 9.4 | 32.1 | 44.6 | 49.7 |
| 30 pm | 12.4 | 9.5 | 28.5 | 44.1 | 48.6 |
| 50 pm | 12.3 | 9.1 | 28.1 | 44.0 | 48.7 |
| 100 pm | 12.2 | 9.2 | 23.1 | 44.4 | 49.5 |
| 150 pm | 11.8 | 8.1 | 21.9 | 44.3 | 45.5 |
| 200 pm | 10.4 | 6.3 | 17.2 | 44.2 | 45.6 |

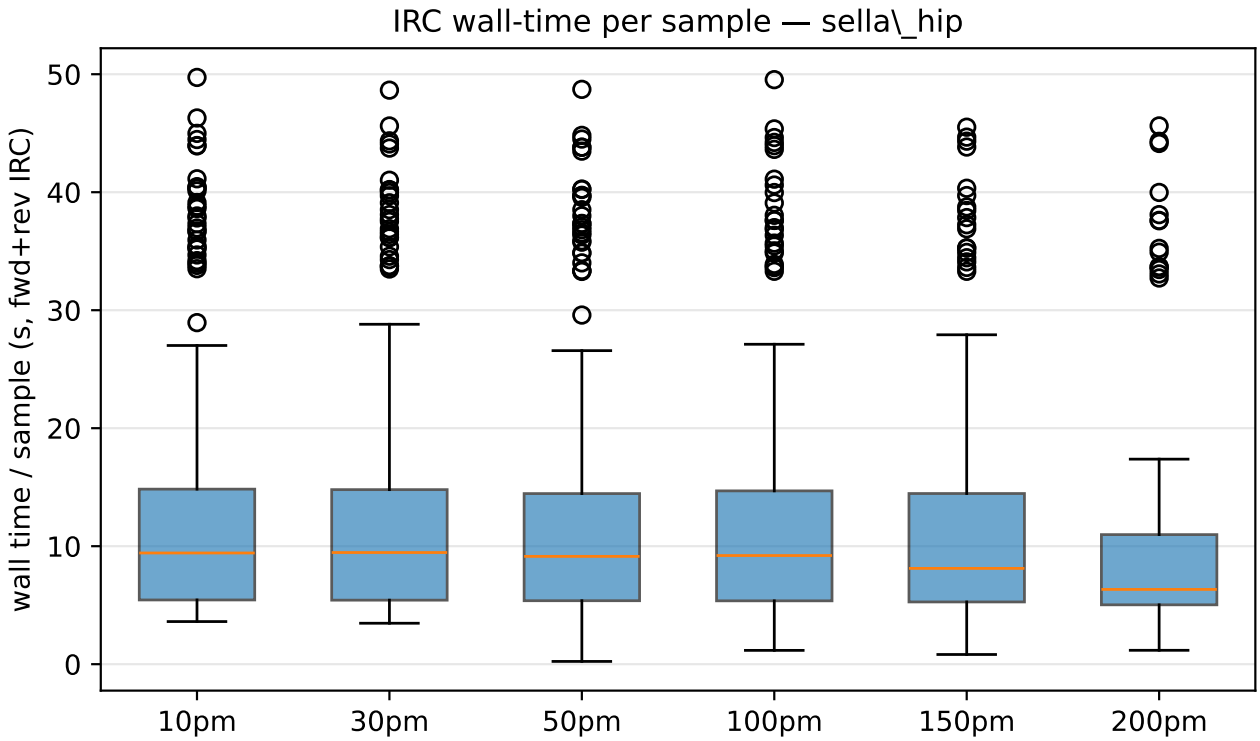


**Figure 6:** IRC wall time per sample (forward+reverse, up to 500 steps each direction). Typical completion is 5–15 s; the long tail reflects hard cases where Sella's inner loops hit the 500-step ceiling.

## Summary and Implications

- **92.1% TOPO-intended across all noise levels, with zero errors.** sella_hip is a robust, production-grade IRC on HIP.
- **Strict RMSD-intended is 64.7%.** The ∼27 pp gap is not a noise artefact — it does not improve at lower noise. It is the natural conformational spread of the labeled endpoints.
- **Only 0.5–1.8% of runs end on a ridge** (neither endpoint at a vibrational minimum). Integrator reliability is excellent.
- **5.8% are "valid-but-wrong" minima** — IRC reached real minima, just not the labeled ones.

Eight out of the ten persistent-failure samples are this case. This is a dataset labeling / reaction-channel ambiguity, not a method failure.

- **Degradation only kicks in at 200 pm**, and even there TOPO-intended is >80%.
- Failing samples are spread across $|\lambda_0|$ and cstep at the TS side — there is no single easy filter that would predict IRC-failure from GAD-side telemetry.
- **Wall time is modest:** median ∼9 s per sample across fwd+rev. The full 1462-run sweep finished in ∼30 min wall parallelized over 6 MIG slices.

**Next.** The rigorous predictor-corrector IRC sweep (Hratchian-Schlegel EulerPC-style) is running in parallel (job 59464202, ETA ∼11 h). Direct head-to-head against sella_hip on the same $n = 1462$ TS set.